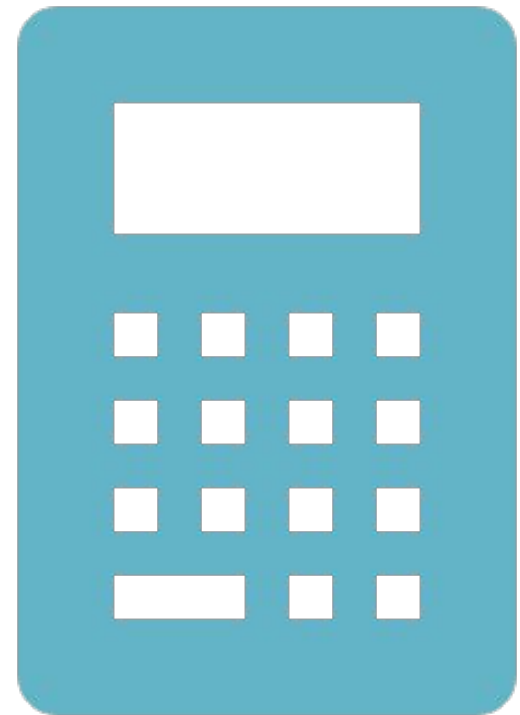


NUMBER SYSTEMS



DECIMAL NUMBERS

- Think about the numbers, you learned at school while growing up.

- They had 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- The number 1,248 is

$$1,000 + 200 + 40 + 8$$

- We can change the representation to:

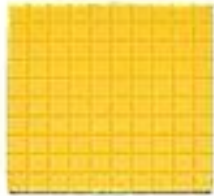
$$(1 * 1,000) + (2 * 100) + (4 * 10) + (8 * 1)$$

- We can change it even further to:

$$(1 * 10^3) + (2 * 10^2) + (4 * 10^1) + (8 * 10^0)$$

DECIMAL NUMBERS (CONT.)

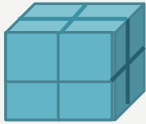
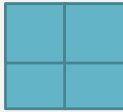
Place Value Chart

			
thousands	hundreds	tens	ones
• 10^3	10^2	10^1	10^0

_____ + _____ + _____ + _____

BINARY

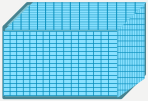
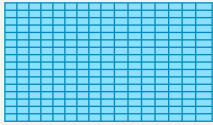
- Digits: 0, 1
- Base 2
- *binary* digit $\square$ *bit*

_____ + _____ + _____ + _____

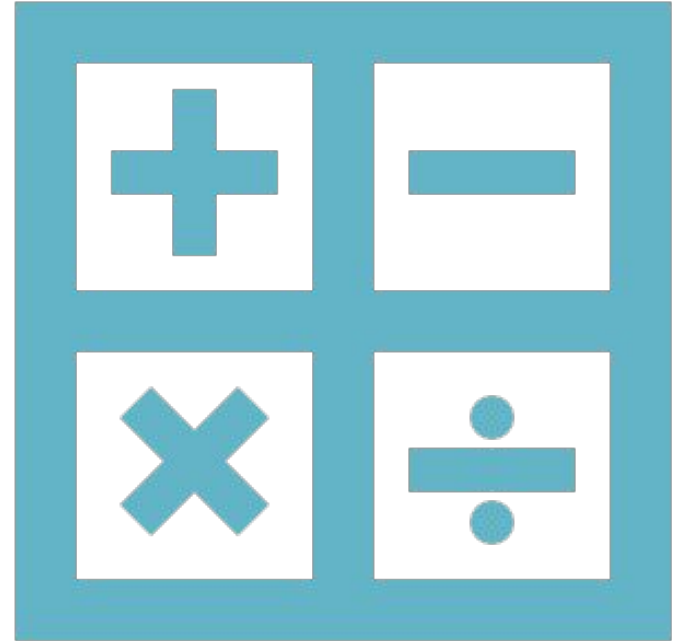
HEXADECIMAL

- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Base 16

_____ + _____ + _____ + _____

CONVERSIONS



Decimal (Base 10)	Binary (Base 2)	Hexadecimal (Base 16)
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

CONVERTING FROM DECIMAL

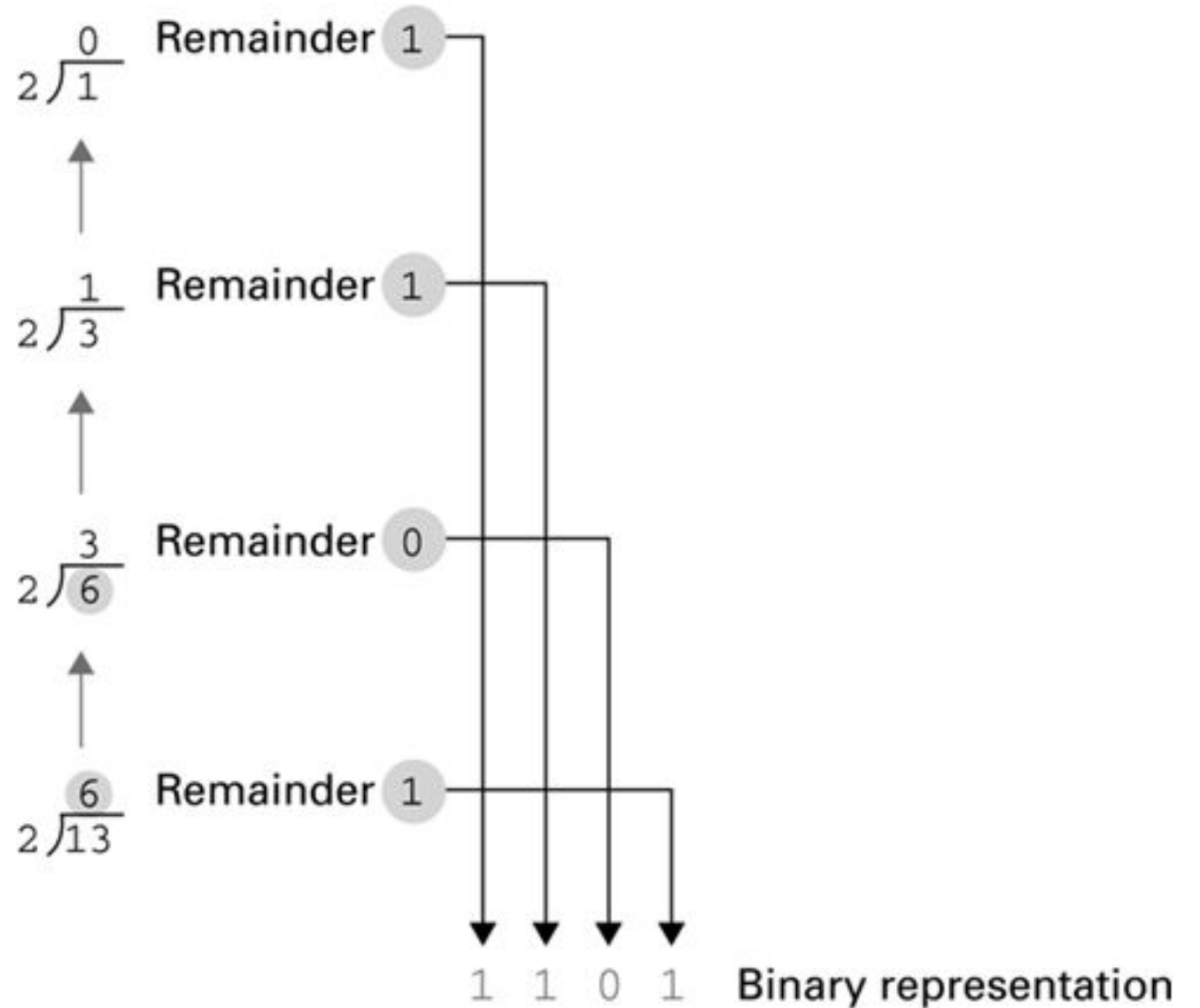
Step 1. Divide the value by the desired base number (B) and record the remainder.

Step 2. As long as the quotient obtained is not zero, continue to divide the newest quotient by B and record the remainder.

Step 3. Now that a quotient of zero has been obtained, the new number system representation of the original value consists of the remainders listed from right to left in the order they were recorded.

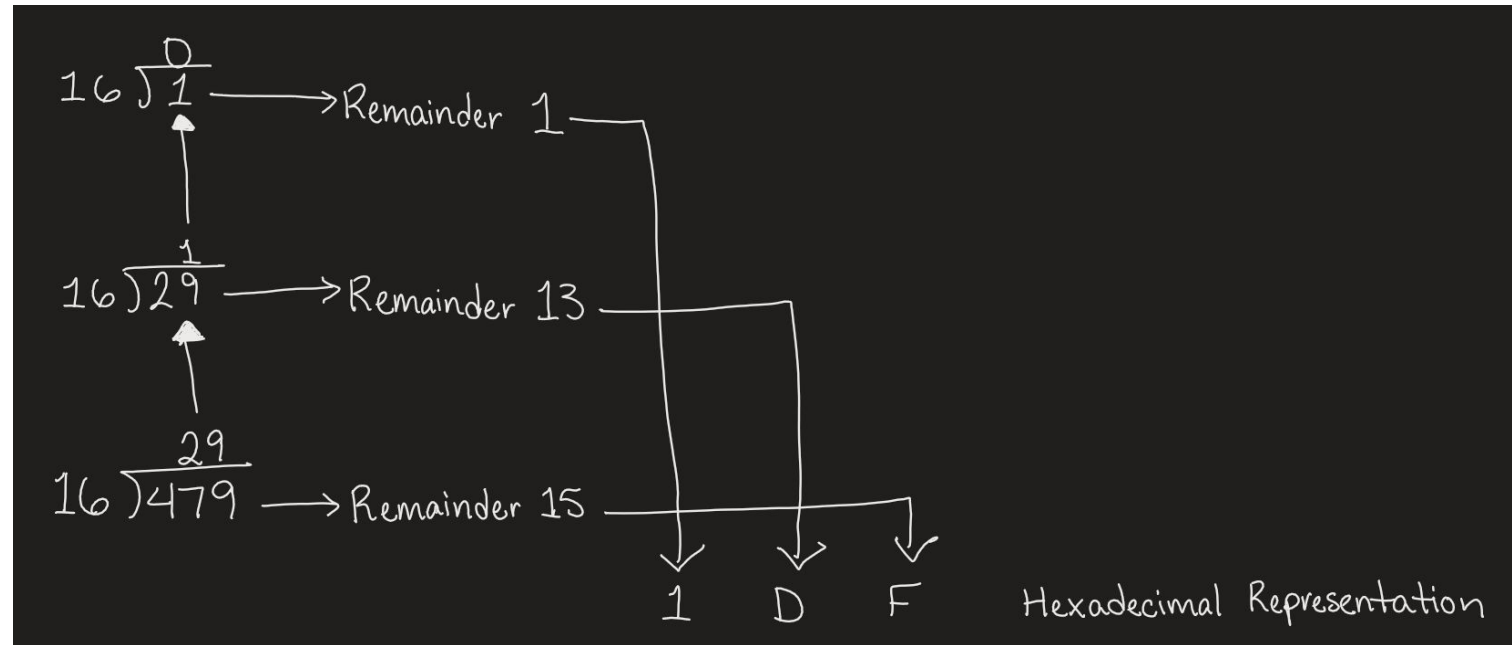
EXAMPLE 1-1

START HERE □



EXAMPLE 1-2

START HERE □

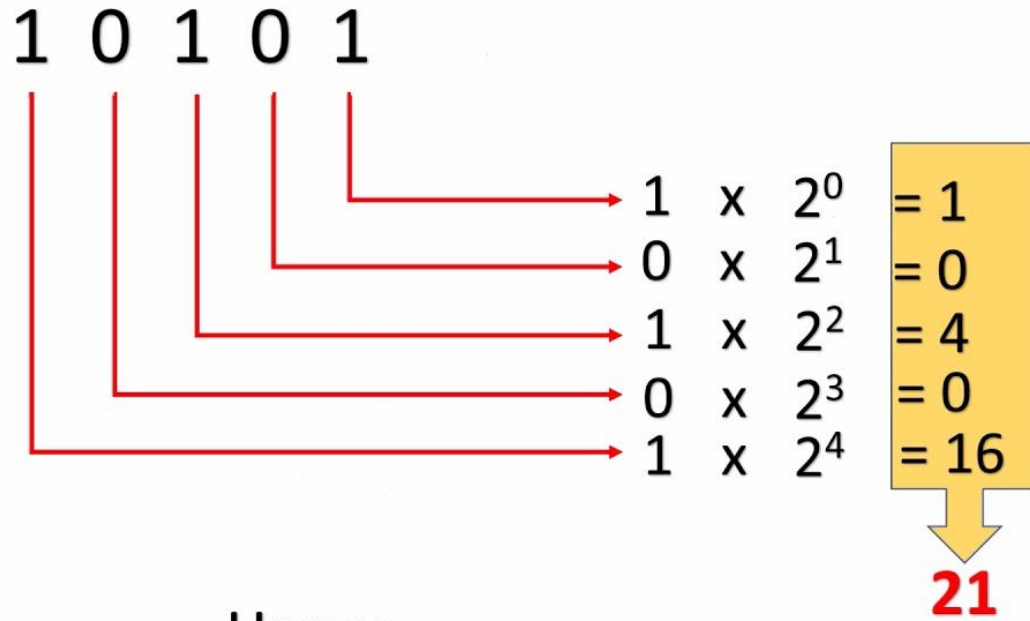


CONVERTING TO DECIMAL

Step 1. Multiply the digit value to quantity it represents (B^n) and record the value. (Recommended to start from right end.)

Step 2. Repeat step 1 until no digits in the number given are left.

Step 3. Add all the recorded values together to obtain the equivalent decimal value.



Hence,

$$10101_2 = 21_{10}$$

EXAMPLE 2-1

A	2	F	7
----------	----------	----------	----------

16^3

16^2

16^1

16^0

decimal:

$$\begin{array}{rcl}
 & & \text{7} \times 16^0 = \text{7} \\
 & & \text{15} \times 16^1 = \text{240} \\
 & & \text{2} \times 16^2 = \text{512} \\
 & & \text{10} \times 16^3 = \text{40960}
 \end{array}$$

41719

EXAMPLE 2-2

CONVERTING FROM BINARY TO HEXADECIMAL

Step 1. Mark groups of four bits (starting at the right end)

Step 2. Convert each group of bits into hex digits.

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

10101011

1010

1011

A

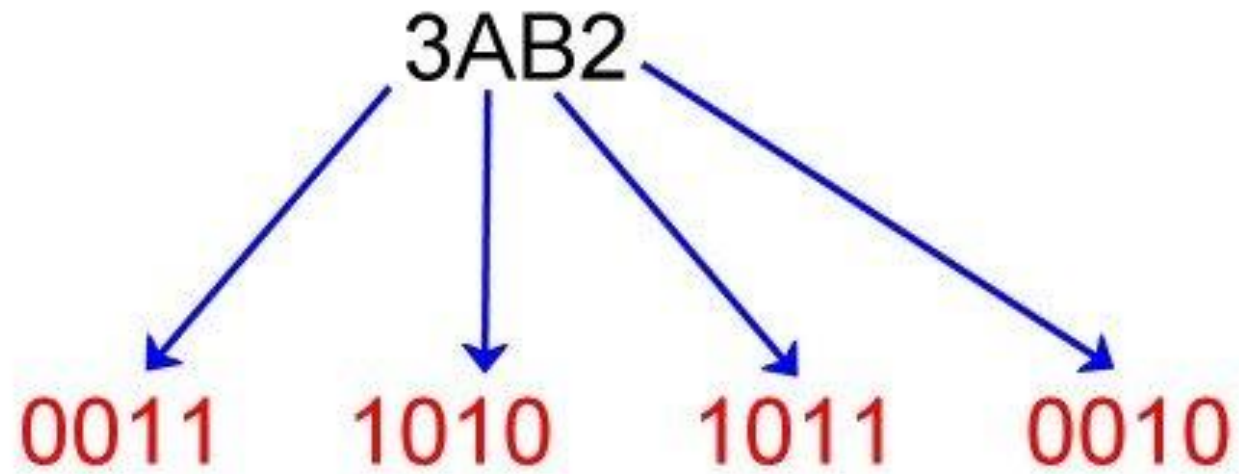
B

EXAMPLE 3

CONVERTING FROM HEXADECIMAL TO BINARY

Step 1. Convert each hex digit into binary from left to right.

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F



$$3AB2_{16} = 11101010110010_2$$

EXAMPLE 4

REFERENCES



- <https://i.pinimg.com/originals/68/c1/96/68c196856e2a8c9ab9edd287300ff48d.jpg>
- <https://i.ytimg.com/vi/4M6L-ubixJo/maxresdefault.jpg>
- https://www.cplusplus.com/doc/hex/base_hexadecimal.gif
- https://usercontent2.hubstatic.com/14086459_f520.jpg
- https://cdn.ttgtmedia.com/rms/onlinelimages/storage-memory_capacity_chart_mobile.png